

Exp. No. 21

Write a LEX specification file to take input C program from a .c file and count the number of characters, number of lines & number of words.

Input Source Program: (sample.c)

```
#include <stdio.h>
int main()
{
    int number1, number2, sum;
    printf("Enter two integers: ");
    scanf("%d %d", &number1, &number2);
    sum = number1 + number2;
    printf("%d + %d = %d", number1, number2, sum);
    return 0;
}
```

Program: (count_lines.l)

```
%{
int nchar, nword, nline;
}%
%%
\n { nline++; nchar++; }
[^ \t\n]+ { nword++; nchar += yyleng; }
. { nchar++; }
%%
int yywrap(void) {
return 1;
}
int main(int argc, char *argv[]) {
yyin = fopen(argv[1], "r");
yylex();
printf("Number of characters = %d\n", nchar);
printf("Number of words = %d\n", nword);
printf("Number of lines = %d\n", nline);
fclose(yyin);
}
```

Output:

```
C:\Users\Administrator\Documents\exp 21.exe

G:\lex>flex count_line.1
G:\lex>gcc lex.yy.c
G:\lex>a.exe sample.c
Number of characters = 233
Number of words = 33
Number of lines = 10

G:\lex>_
```

Exp. No. 22

Write a LEX program to print all the constants in the given C source program file.

Input Source Program: (sample.c)

```
#define P 314
#include<stdio.h>
#include<conio.h>
void main()
{

    int a,b,c = 30;
    printf("hello");
}
```

Program: (countconstants.l)

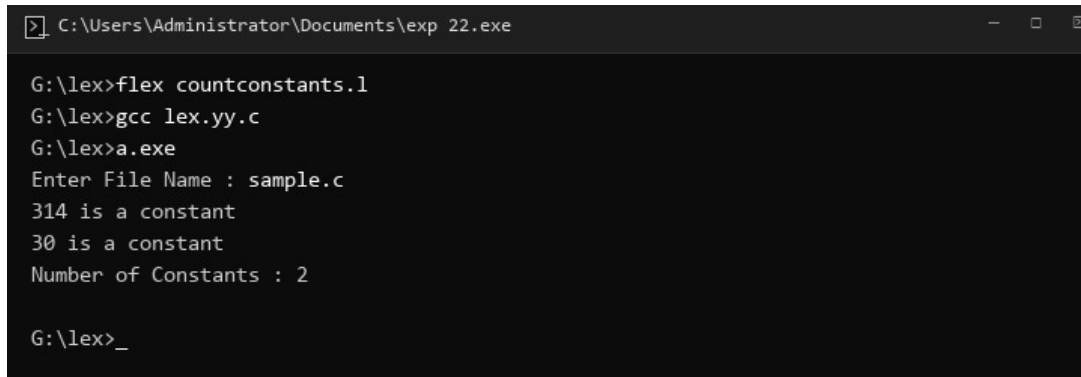
```
digit [0-9]
%{
int cons=0;
}%
%%
{digit}+ { cons++; printf("%s is a constant\n", yytext); }
.\n { }
%%
int yywrap(void) {
return 1; }
int main(void)
{
FILE *f;
char file[10];
printf("Enter File Name : ");
scanf("%s",file);
```

```

f = fopen(file, "r");
yyin = f;
yylex();
printf("Number of Constants : %d\n", cons);
fclose(yyin);
}

```

OUTPUT :



```

C:\Users\Administrator\Documents\exp 22.exe
G:\lex>flex countconstants.l
G:\lex>gcc lex.yy.c
G:\lex>a.exe
Enter File Name : sample.c
314 is a constant
30 is a constant
Number of Constants : 2

G:\lex>_

```

Exp. No. 23

Write a LEX program to count the number of Macros defined and header files included in the C program.

Input Source Program: (sample.c)

```

#define PI 3.14
#include<stdio.h>
#include<conio.h>
void main()
{

    int a,b,c = 30;
    printf("hello");
}

```

Program: (count_macro.l)

```

%{
int nmacro, nheader;
}%
%%
^#define { nmacro++; }
^#include { nheader++; }
.\n { }

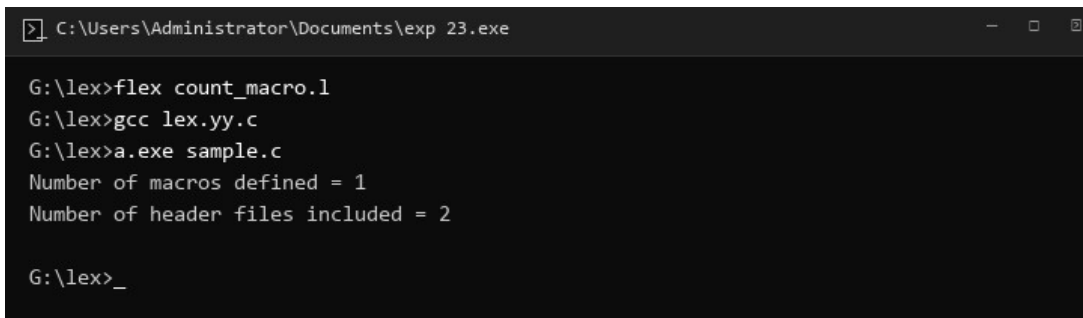
```

```

%%
int yywrap(void) {
return 1;
}
int main(int argc, char *argv[]) {
yyin = fopen(argv[1], "r");
yylex();
printf("Number of macros defined = %d\n", nmacro);
printf("Number of header files included = %d\n", nheader);
fclose(yyin);
}

```

Output:



```

C:\Users\Administrator\Documents\exp 23.exe
G:\lex>flex count_macro.l
G:\lex>gcc lex.yy.c
G:\lex>a.exe sample.c
Number of macros defined = 1
Number of header files included = 2
G:\lex>_

```

Exp. No. 24

Write a LEX program to print all HTML tags in the input file.

Input Source Program: (sample.html)

```

<html>
<body>
<h1>My First Heading</h1>
<p>My first paragraph.</p>
</body>
</html>

```

Program: (html.l)

```

%{
int tags;
}%
%%
"<"[^>]*> { tags++; printf("%s \n", yytext); }
.\n { }
%%

```

```

int yywrap(void) {
return 1; }
int main(void)
{
FILE *f;
char file[10];
printf("Enter File Name : ");
scanf("%s",file);
f = fopen(file,"r");
yyin = f;
yylex();
printf("\n Number of html tags: %d",tags);
fclose(yyin);
}

```

Output:

```

C:\Users\Administrator\Documents\exp 24.exe

G:\lex>flex html.l
G:\lex>gcc lex.yy.c
G:\lex>a.exe
Enter File Name : sample.html
<html>
<body>
<h1>
</h1>
<p>
</p>
</body>
</html>

Number of html tags: 8
G:\lex>_

```

Exp. No. 25

Write a LEX program which adds line numbers to the given C program file and display the same in the standard output.

Input Source Program: (sample.c)

```

#define PI 3.14
#include<stdio.h>
#include<conio.h>
void main()
{

int a,b,c = 30;

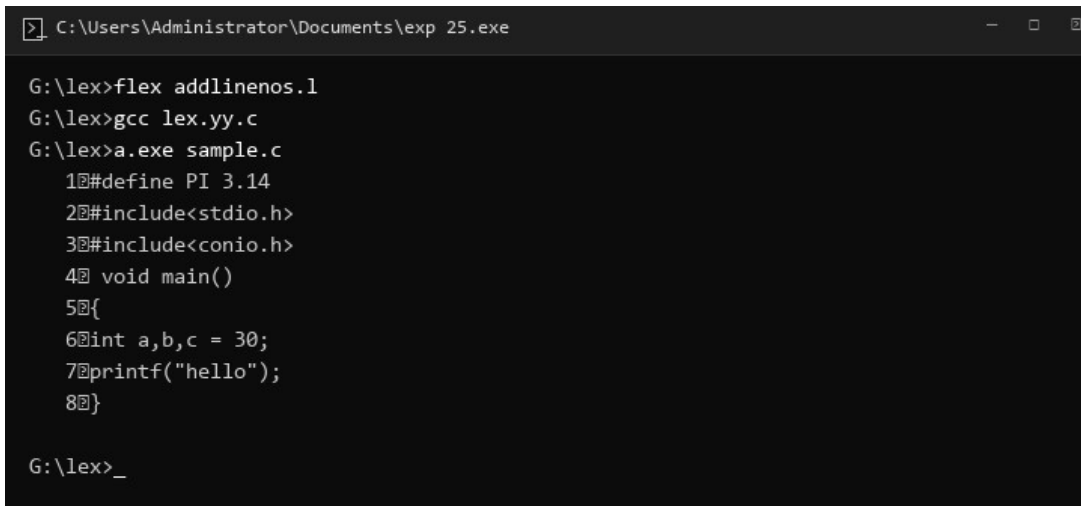
```

```
printf("hello");  
}
```

Program: (addlinenos.l)

```
%{  
int yylineno;  
%}  
%%  
^(.*)\n printf("%4d\t%s", ++yylineno, yytext);  
%%  
int yywrap(void) {  
return 1;  
}  
int main(int argc, char *argv[]) {  
yyin = fopen(argv[1], "r");  
yylex();  
fclose(yyin);  
}
```

Output:



```
C:\Users\Administrator\Documents\exp 25.exe  
  
G:\lex>flex addlinenos.l  
G:\lex>gcc lex.yy.c  
G:\lex>a.exe sample.c  
1#define PI 3.14  
2#include<stdio.h>  
3#include<conio.h>  
4 void main()  
5{  
6int a,b,c = 30;  
7printf("hello");  
8}  
  
G:\lex>_
```

